

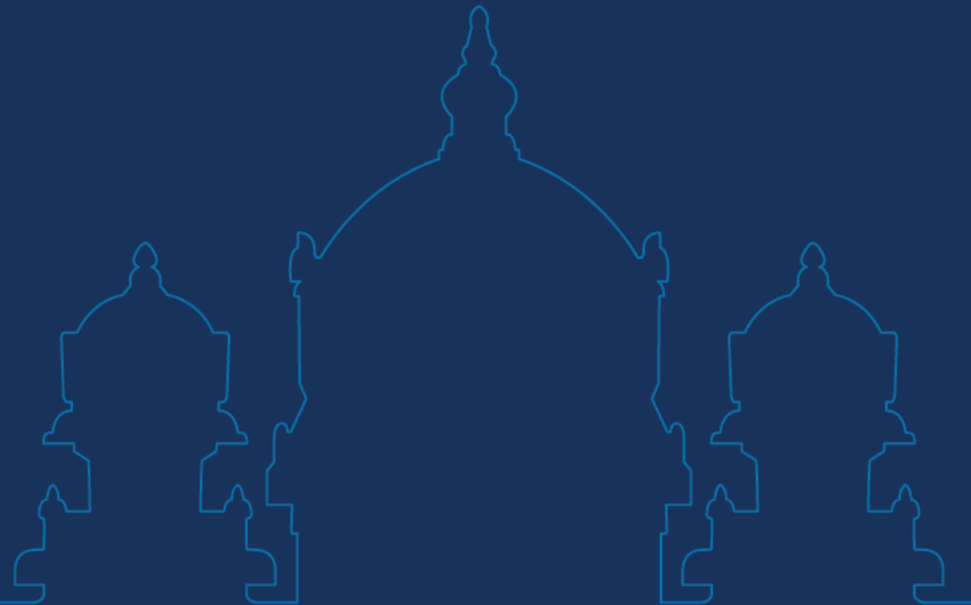
Data Science and Visualization

Unit II DESCRIPTIVE STATISTICS



KCED

Learning Augmented



Unit II – Descriptive Statistics

Advanced Attribute Types - Measures of Central Tendency - Measures of Dispersion - Skewness and Kurtosis - Visual Outlier Identification - Missing Data Visualization - Exploratory Data Analysis (EDA) Techniques - Histograms, KDEs, Box Plots, and Violin Plots.

CO2

Perform Exploratory Data Analysis by applying descriptive statistics and univariate/bivariate visualizations to uncover data distributions, patterns, and relationships, while analyzing data quality through the detection of outliers and exploration of missing value patterns.

Unit II – Descriptive Statistics

Deep Dive into Data/Attribute Types: Nominal-Ordinal-Binary-Numeric -Discrete vs. Continuous-Asymmetric Attributes-Measurement Scales - Basic Statistical Descriptions: Measures of Central Tendency -Mean-Median-Mode - Measures of Dispersion -Range-Quartiles-Variance-Standard Deviation-IQR - Understanding Skewness and Kurtosis -Conceptual - Data Cleaning Context: Visual identification of outliers - Visualizing Missing Data –Types of Missingness- **Exploratory Data Analysis Visualization Techniques-Univariate: Histograms with Kernel Density Estimation-Box Plots -Violin Plots.**

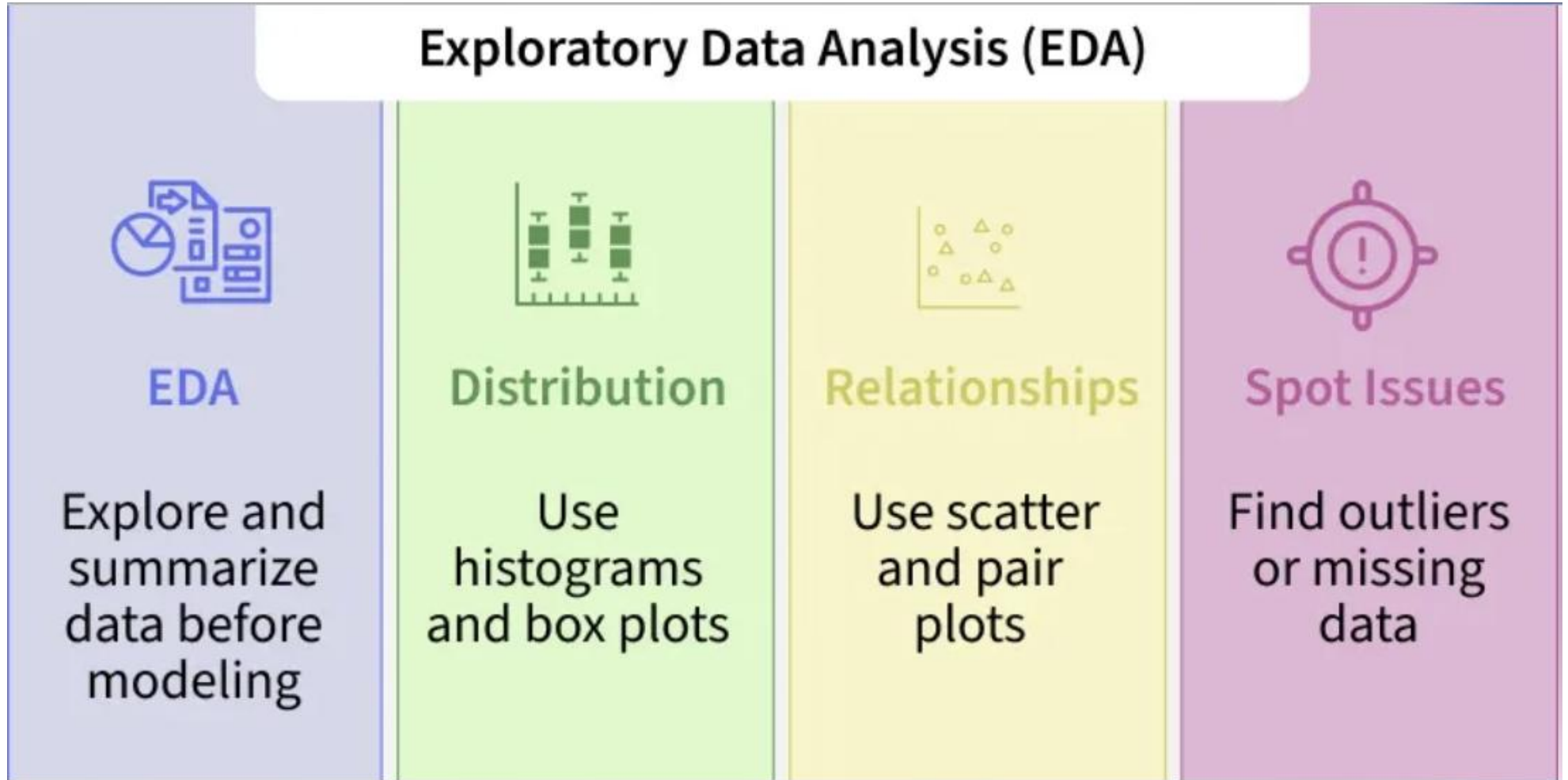
CO2

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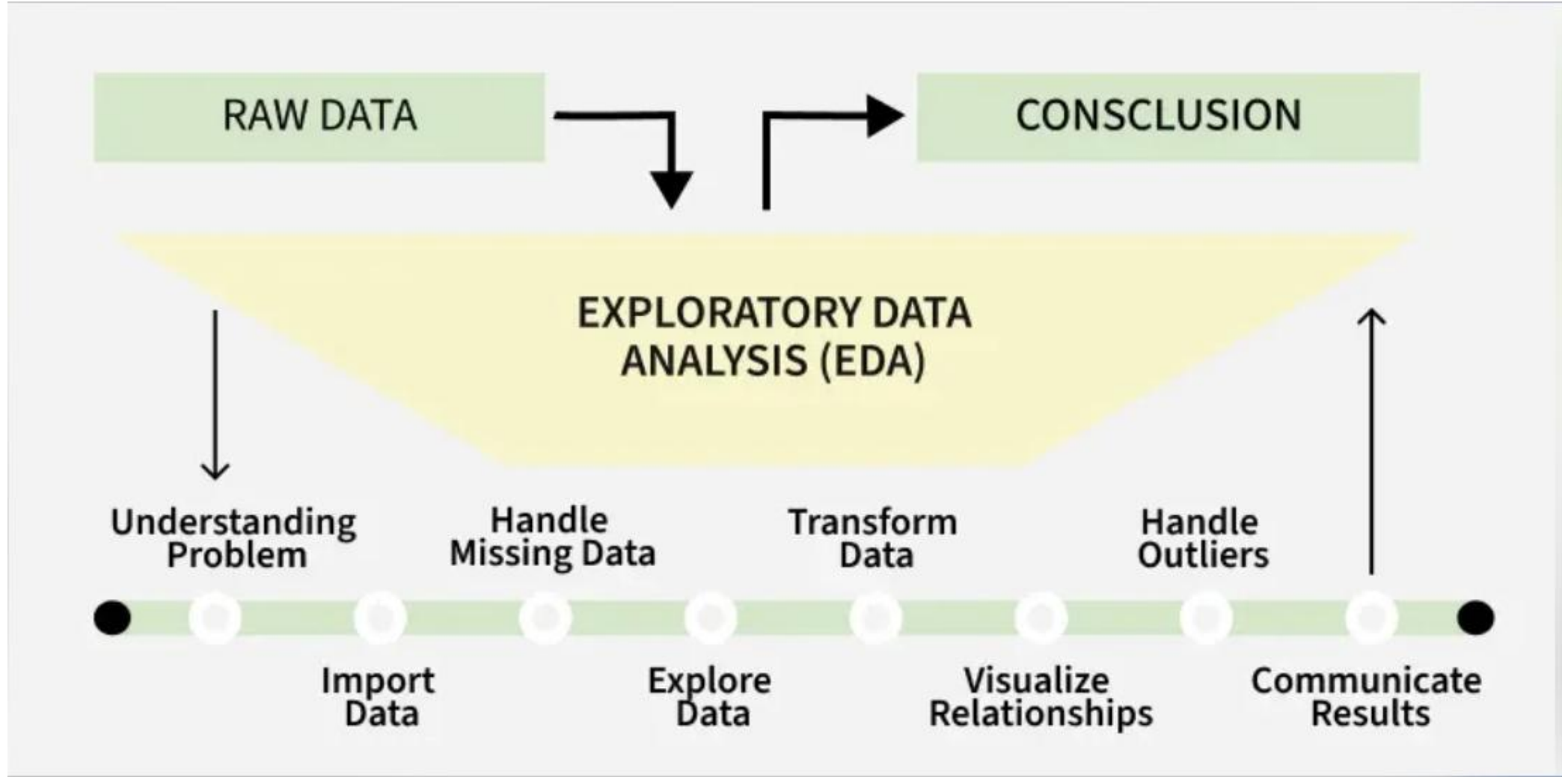
What is EDA?

- Exploratory Data Analysis (EDA) is the **first step in data analysis** used **to understand the structure and characteristics** of data.
- It helps identify **patterns, trends, relationships, and outliers** in the dataset.
- EDA ensures that appropriate statistical and analytical methods are chosen for further analysis.
- Visualization techniques such as **scatterplots, correlation matrices, and multivariate plots** provide insights into variable relationships.

What is EDA?



The EDA Process



The EDA Process

1. **Look at the Data:** Understand the dataset structure - number of rows and columns, data types, and distributions of individual variables.
2. **Clean the Data:** Handle missing, incorrect, or inconsistent values. Preprocessing ensures data quality for analysis and predictive modeling.
3. **Make Summaries:** Use descriptive statistics such as mean, median, mode, quantiles, and skewness to understand overall data characteristics.

Steps in Exploratory Data Analysis (EDA)



Look at the Data

Understand the dataset structure: number of rows and columns, data types, and distributions of individual variables.



Clean the Data

Handle missing, incorrect, or inconsistent values. Preprocessing ensures data quality for analysis and predictive modeling.



Make Summaries

Use descriptive statistics such as mean, median, mode, quantiles, and skewness to understand overall data characteristics.



Visualize the Data

Apply charts and graphs (bar plots, scatter plots, histograms, heatmaps) to identify patterns, trends, and anomalies using `pandas`, `NumPy`, `Matplotlib`, `Seaborn`, `Plotly`.



Ask Questions

Generate meaningful questions based on observations, such as relationships between variables or reasons for unusual values.



Find Answers

Perform deeper analysis or build models (e.g, regression or linear regression) to validate insights and answer questions.

The EDA Process

4. Visualize the Data:

Apply charts and graphs (bar plots, scatter plots, histograms, heatmaps) to identify patterns, trends, and anomalies using tools like pandas, NumPy, Matplotlib, Seaborn, and Plotly.

5. Ask Questions: Generate meaningful questions based on observations, such as relationships between variables or reasons for unusual values.

6. Find Answers: Perform deeper analysis or build models (e.g., regression or linear regression) to validate insights and answer questions.

Steps in Exploratory Data Analysis (EDA)



Look at the Data

Understand the dataset structure: number of rows and columns, data types, and distributions of individual variables.



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Apply charts and graphs (bar plots, scatter plots, histograms, heatmaps) to identify patterns, trends, and anomalies using **pandas**, **NumPy**, **Matplotlib**, **Seaborn**, **Plotly**.



Ask Questions

Generate meaningful questions based on observations, such as relationships between variables or reasons for unusual values.



Find Answers

Perform deeper analysis or build models (e.g, regression or linear regression) to validate insights and answer questions.

The EDA Process

For **example**, in **Python**, you can perform EDA techniques

- by importing necessary libraries,
- loading your dataset, and
- using functions to display basic information,
- summary statistics,
- check for missing values, and
- visualize distributions and relationships between variables.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load the dataset
data = pd.read_csv('your_dataset.csv')

# Display basic information
print(data.info())

# Display summary statistics
print(data.describe())

# Check for missing values
print(data.isnull().sum())

# Visualize distributions
sns.histplot(data['column_name'])
plt.show()
```

Why is Exploratory Data Analysis Important?

- **Generating Inaccurate Models:** Models built on unexamined data can be inaccurate and unreliable.
- **Using Wrong Data:** Without EDA, you might be analyzing or modeling the wrong data, leading to false conclusions.
- **Inefficient Resource Use:** Inefficiently using computational and human resources due to lack of proper data understanding.
- **Improper Data Preparation:** EDA helps in creating the right types of variables, which is critical for effective data preparation.

Types of EDA



Univariate

Analyzes one variable



Bivariate

Studies relationships
between two variables

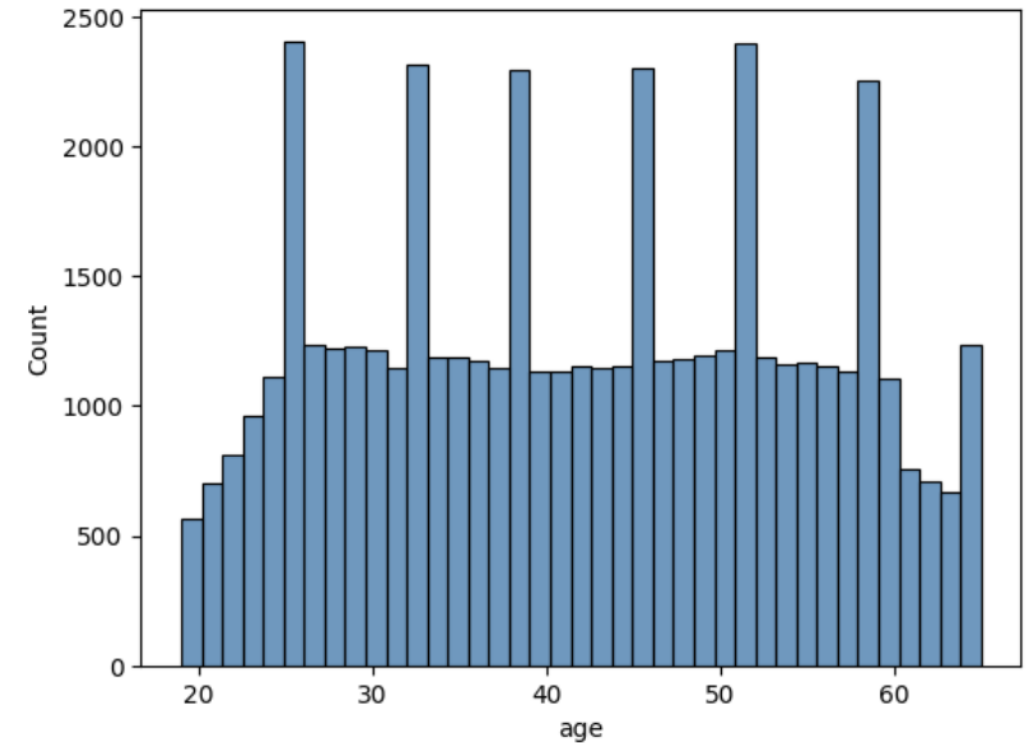


Multivariate

Looks at interactions
between multiple
variables

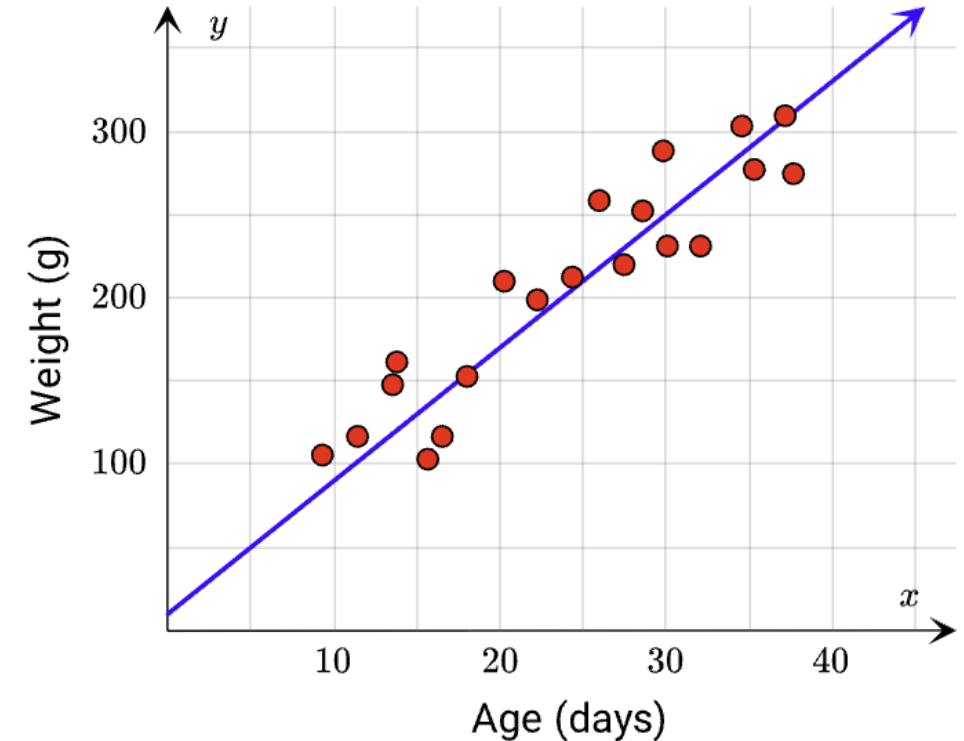
Univariate Analysis

- Univariate analysis **examines individual variables** to understand their distributions and summary statistics.
- This includes calculating measures such as **mean, median, mode, and standard deviation**.
- Visualizing the data using:
 - Histograms
 - Bar charts
 - Box plots
 - Violin plots



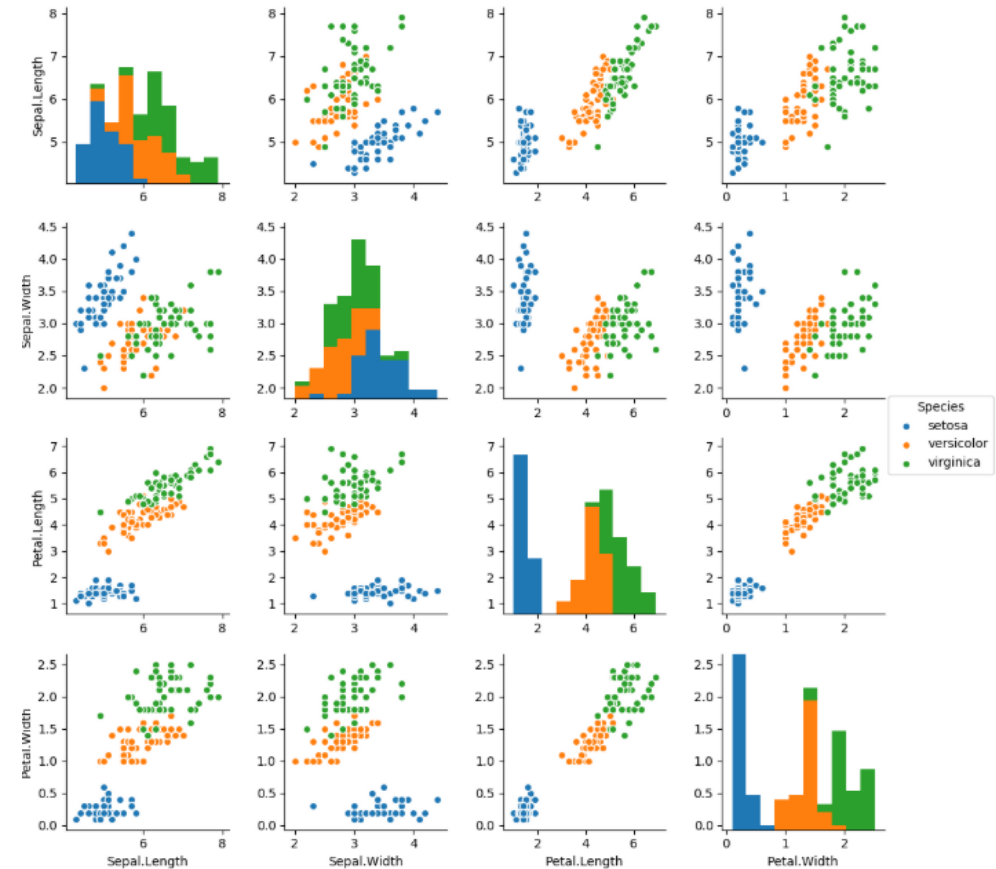
Bivariate Analysis

- Bivariate analysis explores the **relationship between two variables**.
- It uncovers patterns through techniques like scatter plots, pair plots, and heatmaps.
- This helps to **identify potential associations** or **dependencies between variables**.



Multivariate Analysis

- Multivariate analysis involves examining **more than two variables** simultaneously to understand their relationships and combined effects.
- Techniques such as **contour plots**, **pair plots** and **principal component analysis (PCA)** are commonly used in multivariate EDA.

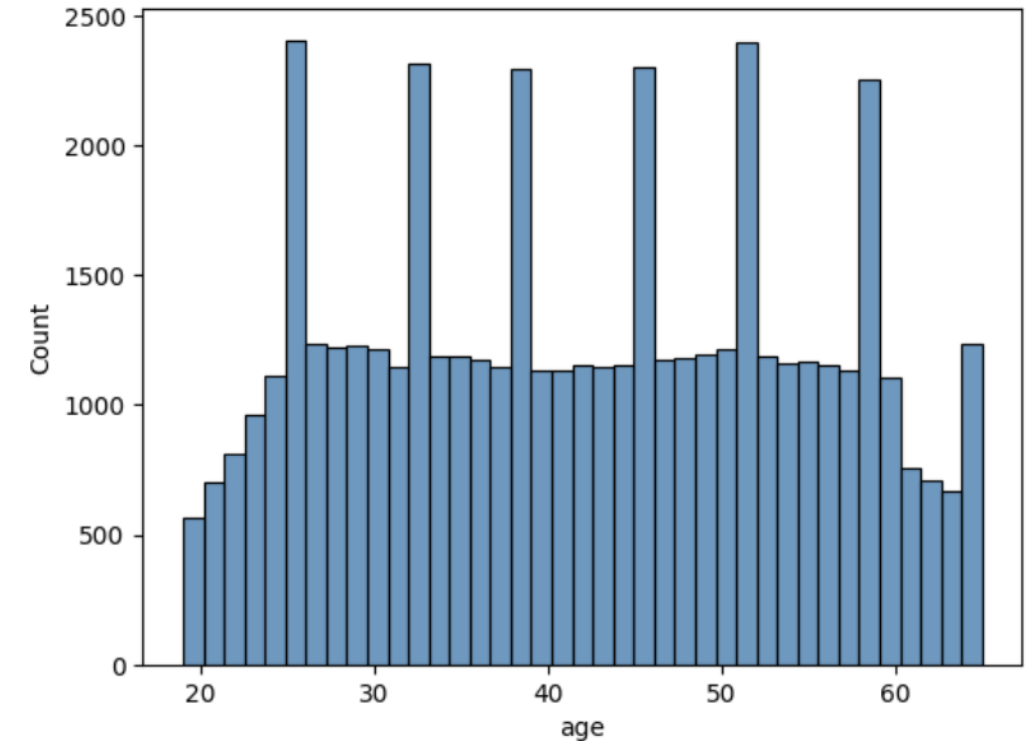


Types of Plots

Plot Type	Category	Purpose
Histogram	Univariate	Distribution analysis
Bar Chart	Univariate	Category comparison
Box Plot	Univariate	Outlier detection
Density Plot	Univariate	Distribution shape
Scatter Plot	Bivariate	Relationship analysis
Line Plot	Bivariate	Trend analysis
Heatmap	Bivariate / Multivariate	Correlation strength
Bubble Chart	Multivariate	3-variable relationship
Pair Plot	Multivariate	Multi-feature analysis
Radar Chart	Multivariate	Multi-metric comparison

Histogram

- A histogram is a **univariate plot** used to represent the **distribution of numerical data** by grouping values into **continuous intervals called bins** and showing their frequency.
- In simple words: It tells us how data is spread across different ranges.



Histogram

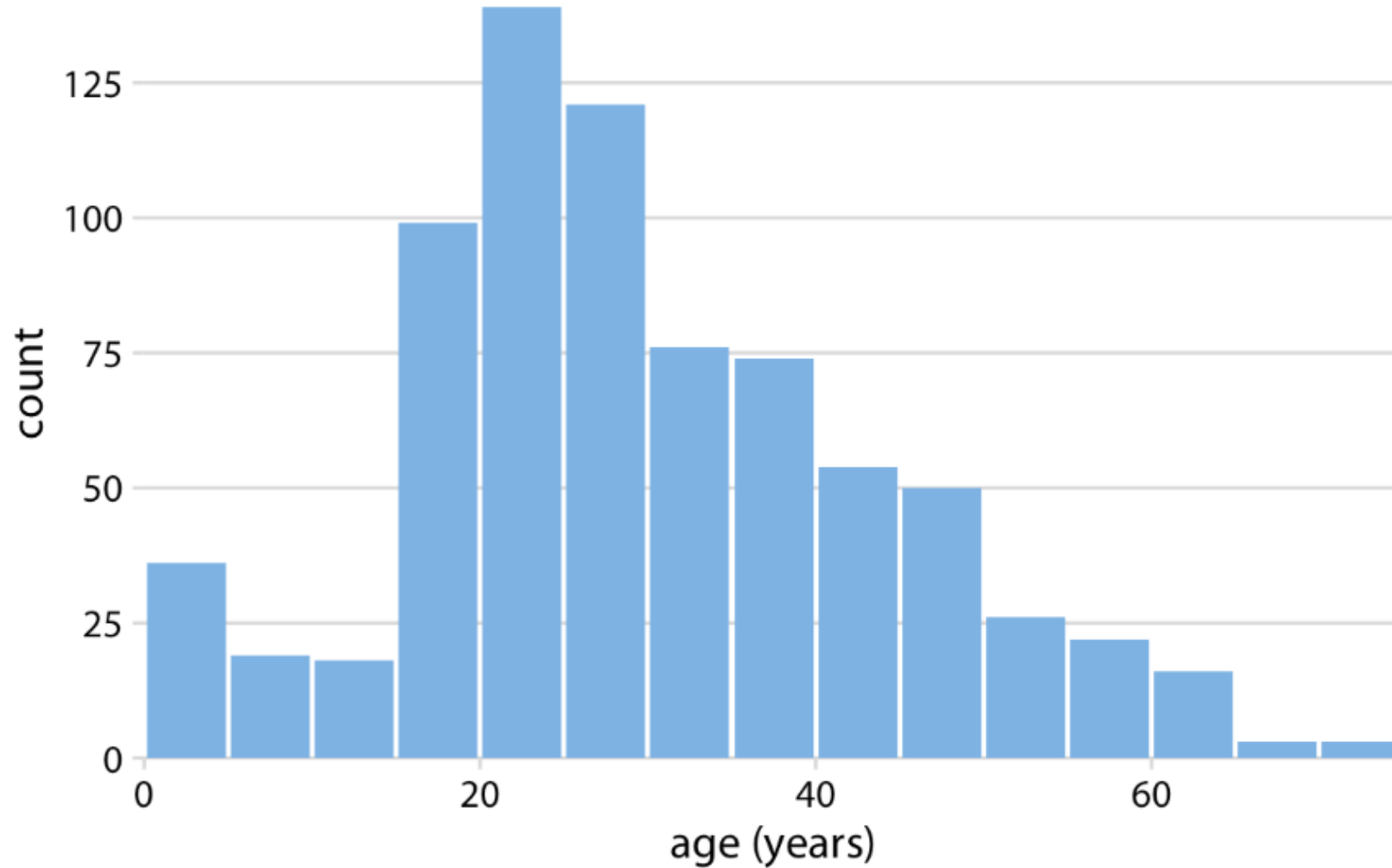
- Histograms are one of the **most fundamental tools** in data visualization.
- They provide a graphical representation of **data distribution**, showing how frequently each value or range of values occurs.
- Histograms are especially useful for **analyzing continuous numerical data**, such as measurements, sensor readings or experimental results.
- A histogram is a **type of bar plot** where:
 - ✓ The **X-axis** represents **intervals (called bins)** of the data.
 - ✓ The **Y-axis** represents the **frequency** of values within each bin.

Visualizing a Single Distribution

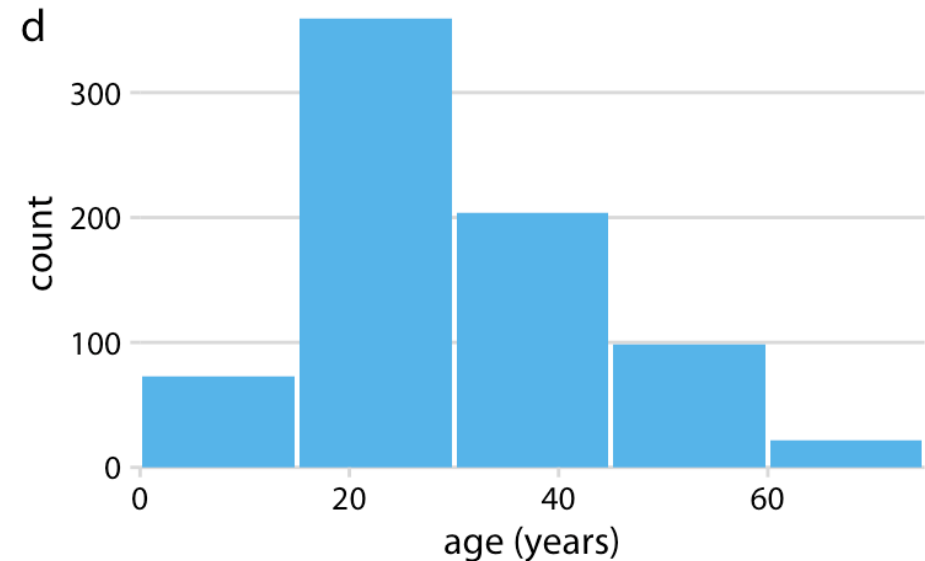
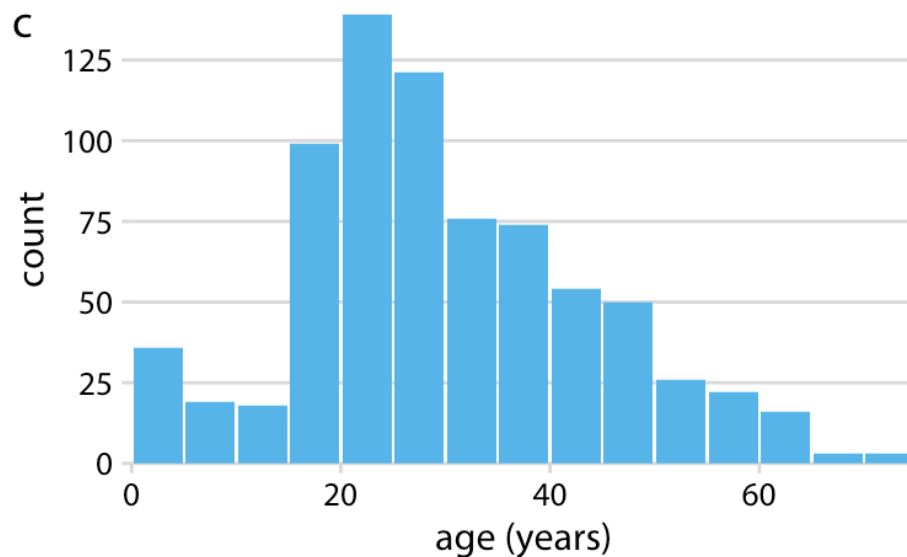
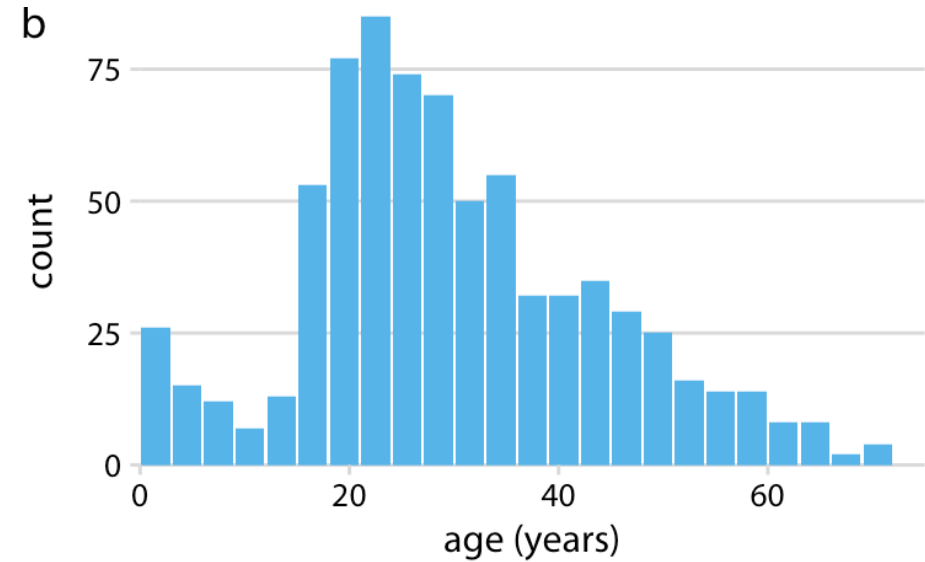
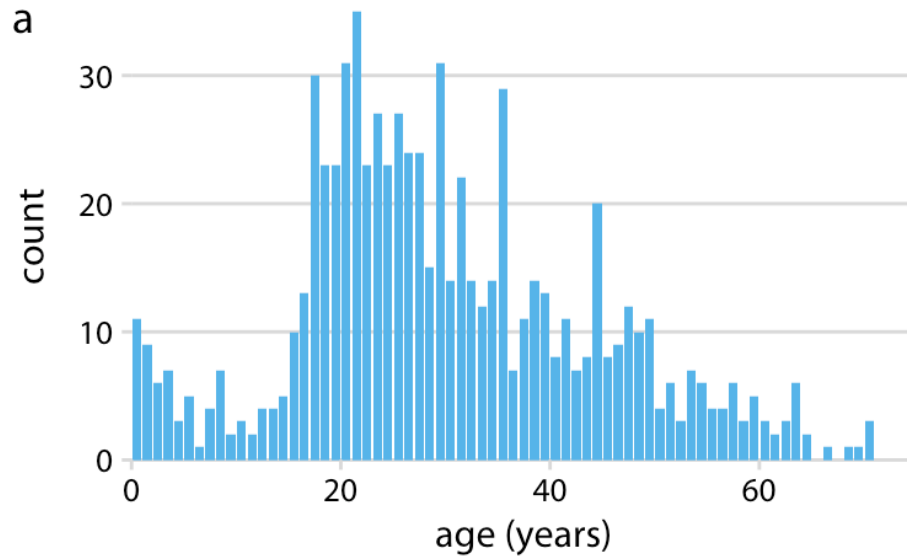
Table 7.1: Numbers of passenger with known age on the Titanic.

Age range	Count	Age range	Count	Age range	Count
0–5	36	31–35	76	61–65	16
6–10	19	36–40	74	66–70	3
11–15	18	41–45	54	71–75	3
16–20	99	46–50	50		
21–25	139	51–55	26		
26–30	121	56–60	22		

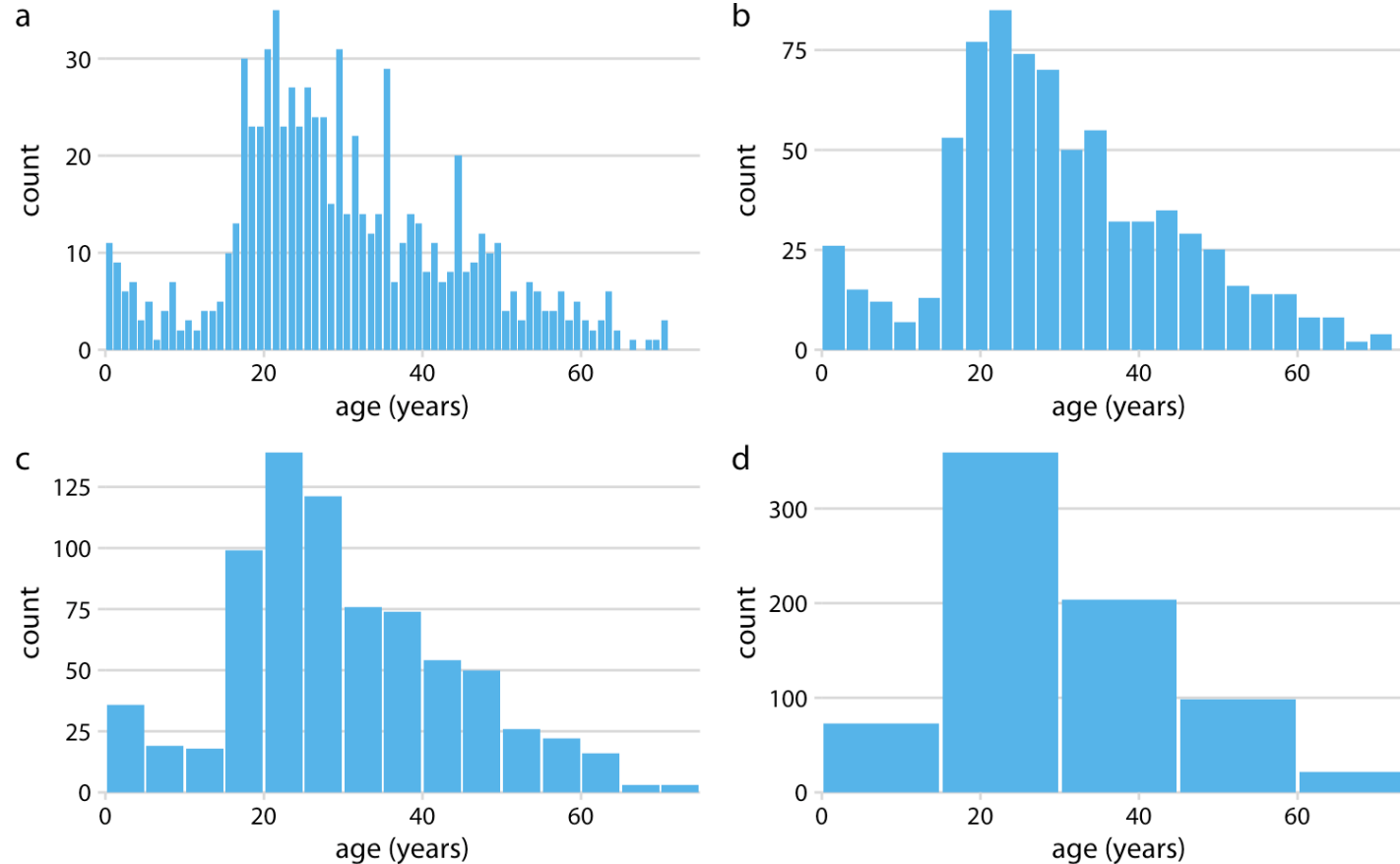
Visualizing a Single Distribution - Histogram



Visualizing a Single Distribution - Histogram

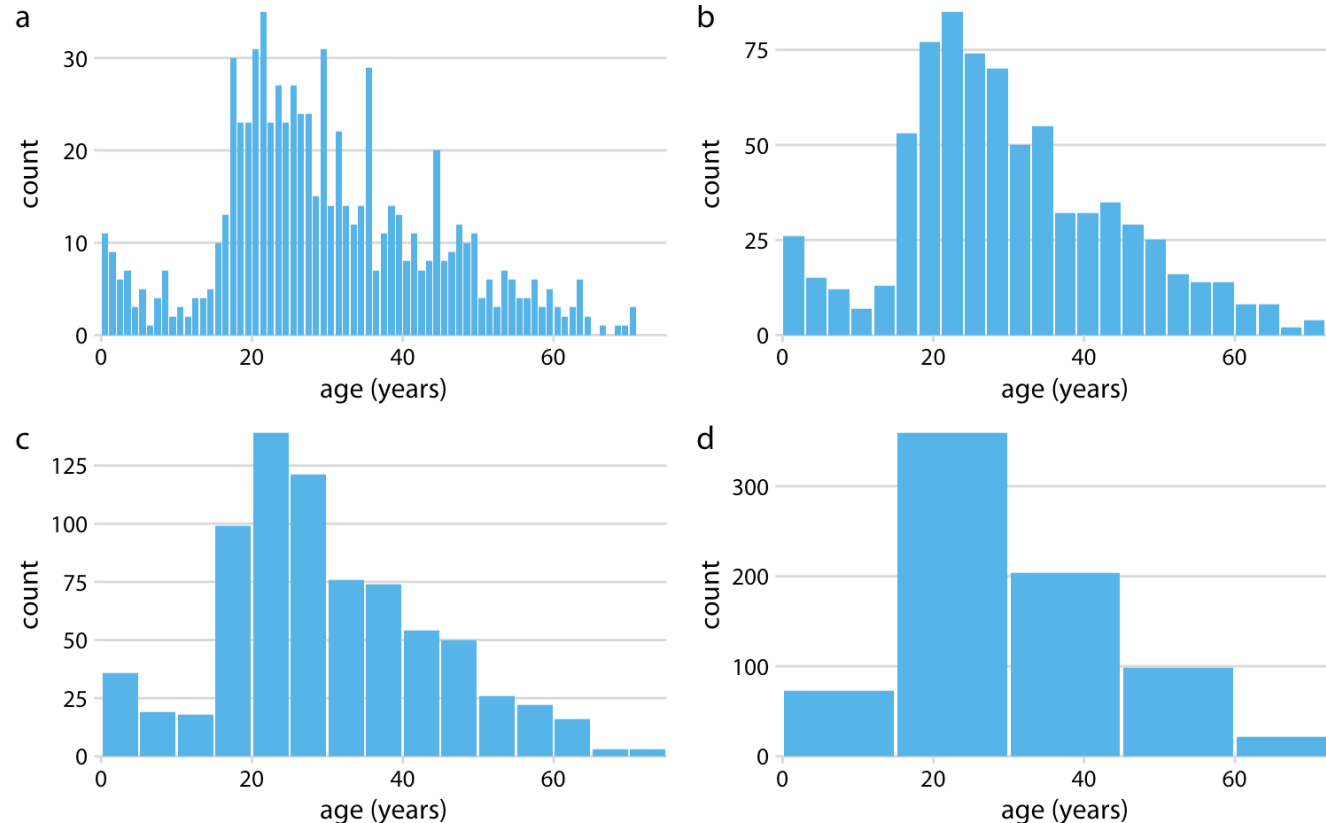


Visualizing a Single Distribution - Histogram



Histograms depend on the chosen bin width. Here, the same age distribution of Titanic passengers is shown with four different bin widths:
(a) one year; (b) three years; (c) five years; (d) fifteen years.

Visualizing a Single Distribution - Histogram



- If the **bin width is too small**, the histogram becomes overly peaky and visually cluttered, making it **difficult to identify the main trends in the data**.
- If the **bin width is too large**, important finer features of the **distribution may be smoothed out and lost**, such as localized dips (for example, around age 10).

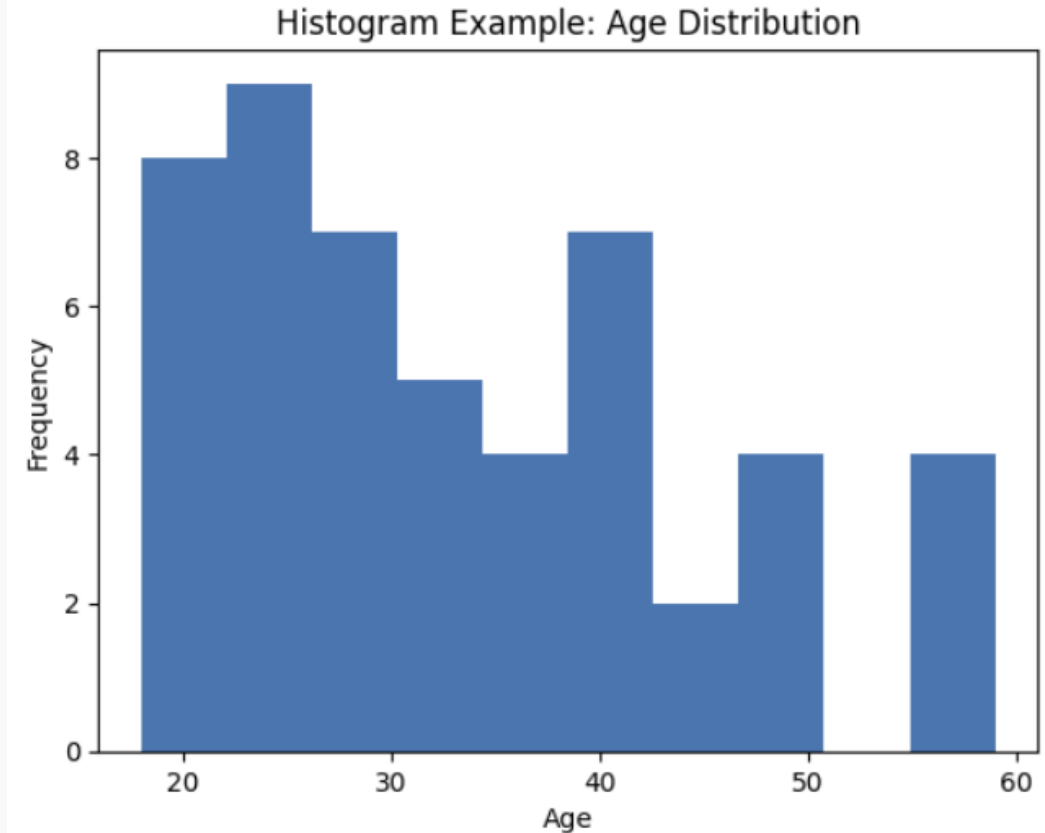
Histogram - Example

Problem: Visualize the distribution of ages of 50 people.

```
import matplotlib.pyplot as plt
import numpy as np

# Sample data
np.random.seed(1)
ages = np.random.randint(18, 60, 50)

# Plot histogram
plt.hist(ages)
plt.title("Histogram Example: Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```

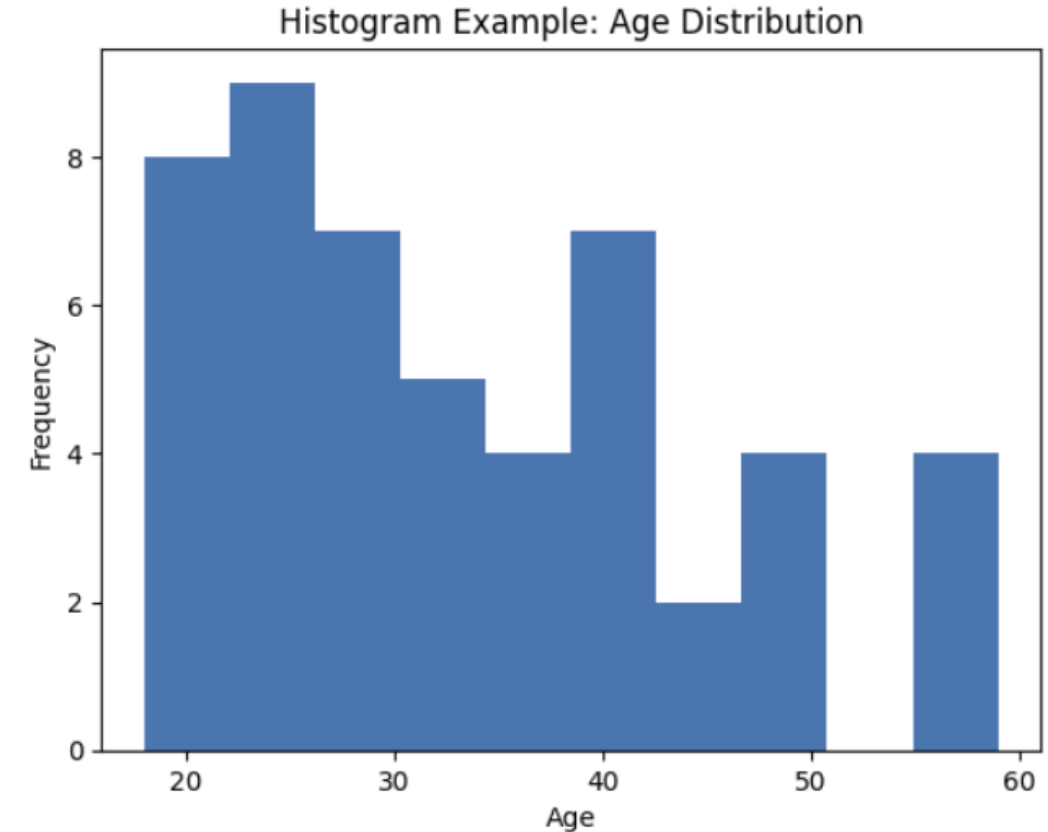


Histogram - Example

Problem: Visualize the distribution of ages of 50 people.

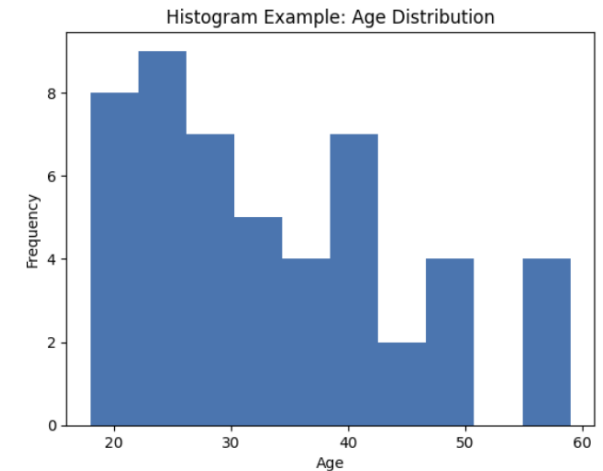
The histogram you see shows:

- **X-axis:** Age groups (bins)
- **Y-axis:** Number of people in each age group
- **Taller bars** → more people in that age range



Where to Use Histogram?

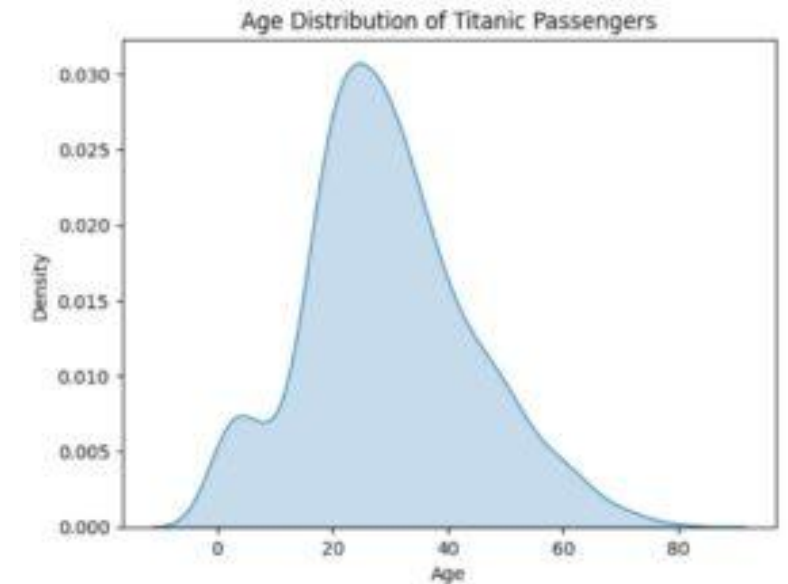
- Visualize the **frequency distribution** of numerical data.
- Used in exploratory data analysis (EDA).
- Helps understand **data spread, shape, and outliers**.
- Useful when exact **frequencies/counts** are important.
- Suitable for **small to moderate datasets**.



KDEs / Density Plots

A KDE (Kernel Density Estimation) plot is a smoothed version of a histogram that estimates the probability density function of a **continuous** variable.

- ✓ It helps to visualize **the shape of data distribution** without being affected by the choice of bin size (as in histograms).
- ✓ It's useful for **identifying peaks, skewness, and multimodal distributions**.



KDEs / Density Plots

- In a density plot, we attempt to visualize the underlying probability distribution of the data by drawing an **appropriate continuous curve**.
- This curve needs to be estimated from the data, and the most commonly used method for this estimation procedure is called **kernel density estimation**.
- In kernel density estimation, we draw a **continuous curve (the kernel)** with a small width (controlled by a parameter called **bandwidth**) at the location of each data point, and then we add up all these curves to obtain the final density estimate.
- The most widely used kernel is a **Gaussian kernel** (i.e., a Gaussian bell curve), but there are many other choices.

KDEs / Density Plots

What is a Kernel?

A **Kernel** is a mathematical function used in **Kernel Density Estimation (KDE)** to smooth data distributions. It helps in estimating the probability density function (PDF) of a **continuous variable**.

◆ How Does a Kernel Work?

- Instead of using **discrete bins** (like a histogram), KDE places a **smooth curve** (Kernel function) over each data point.
- These curves are then **summed together** to create a smooth **density estimate**.
- The **width of the curve** is controlled by the **bandwidth** parameter, which determines how much smoothing is applied.

KDEs / Density Plots

What is Density?

Density in statistics and data visualization refers to how **concentrated** or **spread out** the values of a variable are within a given range. It represents the **relative frequency** of data points in different areas of the variable's distribution.

◆ Understanding Density in Simple Terms

Imagine you have 100 people and you want to analyze their ages.

- If most people are between 20 and 30 years old, this range has a **higher density** of data points.
- If only a few people are between 60 and 70 years old, this range has a **lower density**.

Density helps us understand **which values occur more often** in a dataset.

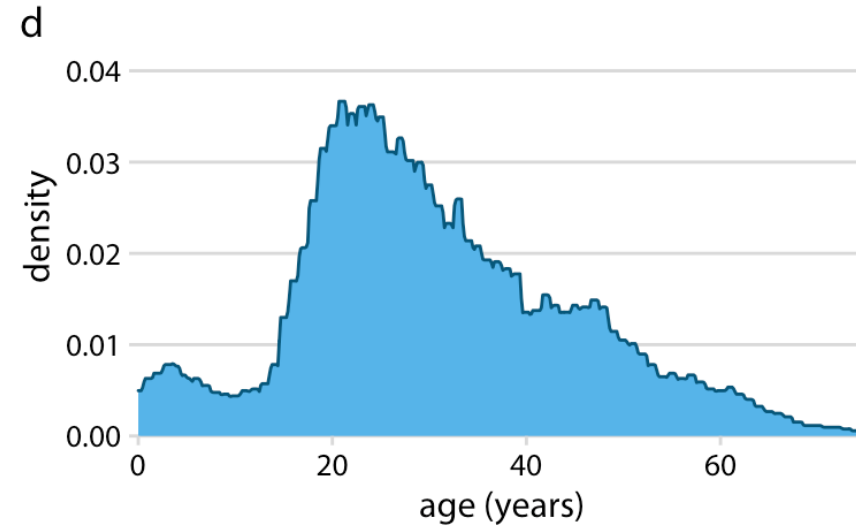
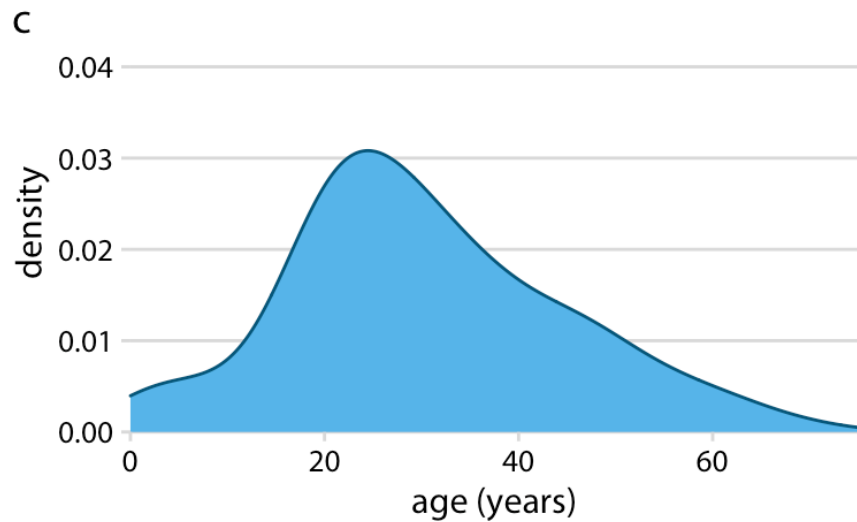
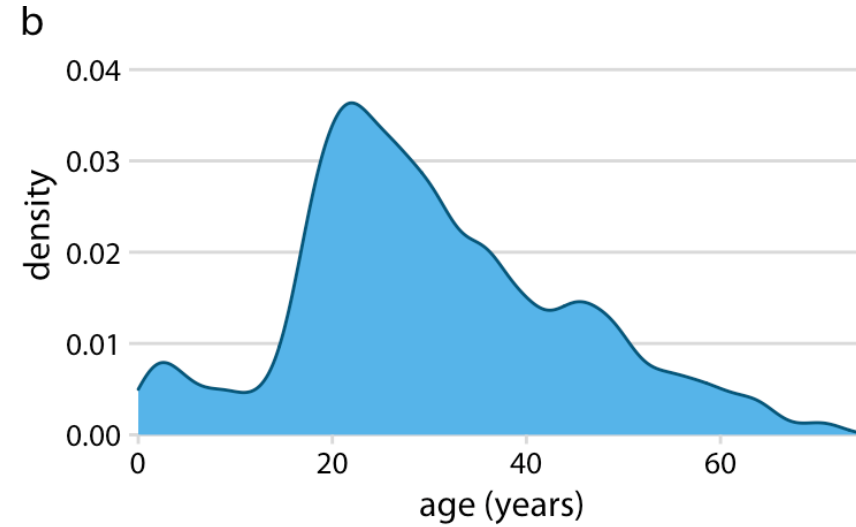
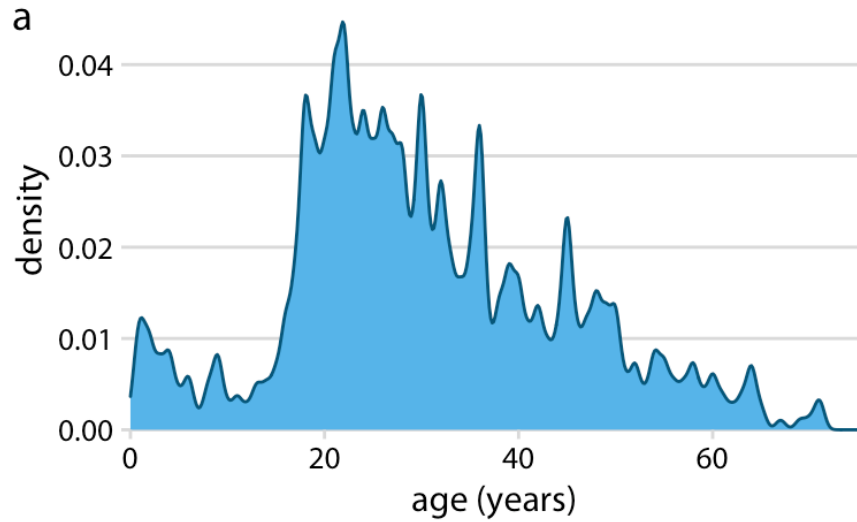
KDEs / Density Plots

◆ Density in KDE (Kernel Density Estimation)

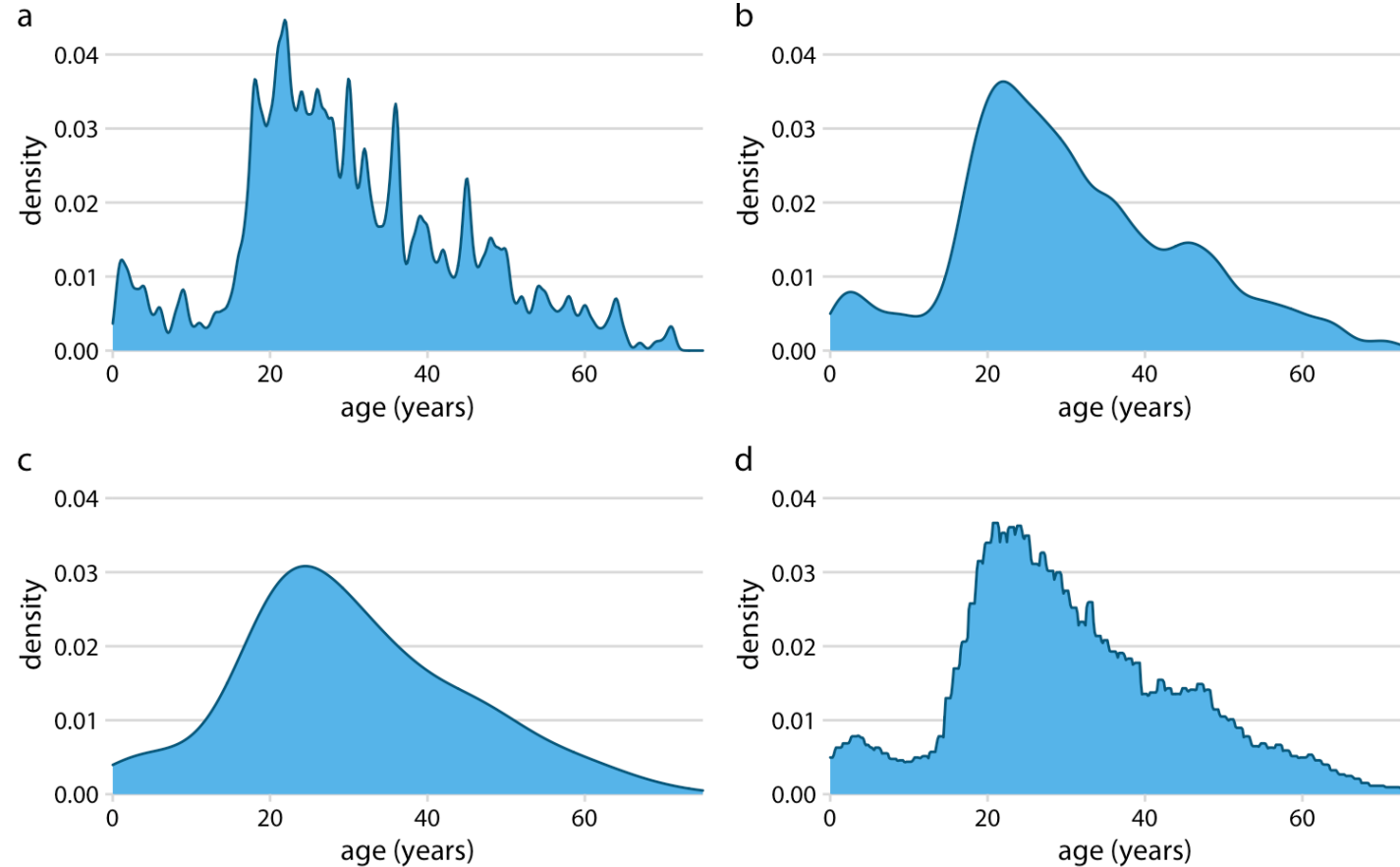
In a KDE plot (`sns.kdeplot()`), density is estimated using a **smooth curve** rather than counting values in fixed bins (as in a histogram).

- **Higher peaks** indicate areas where more data points exist (higher density).
- **Flatter regions** indicate fewer data points (lower density).

KDEs / Density Plots

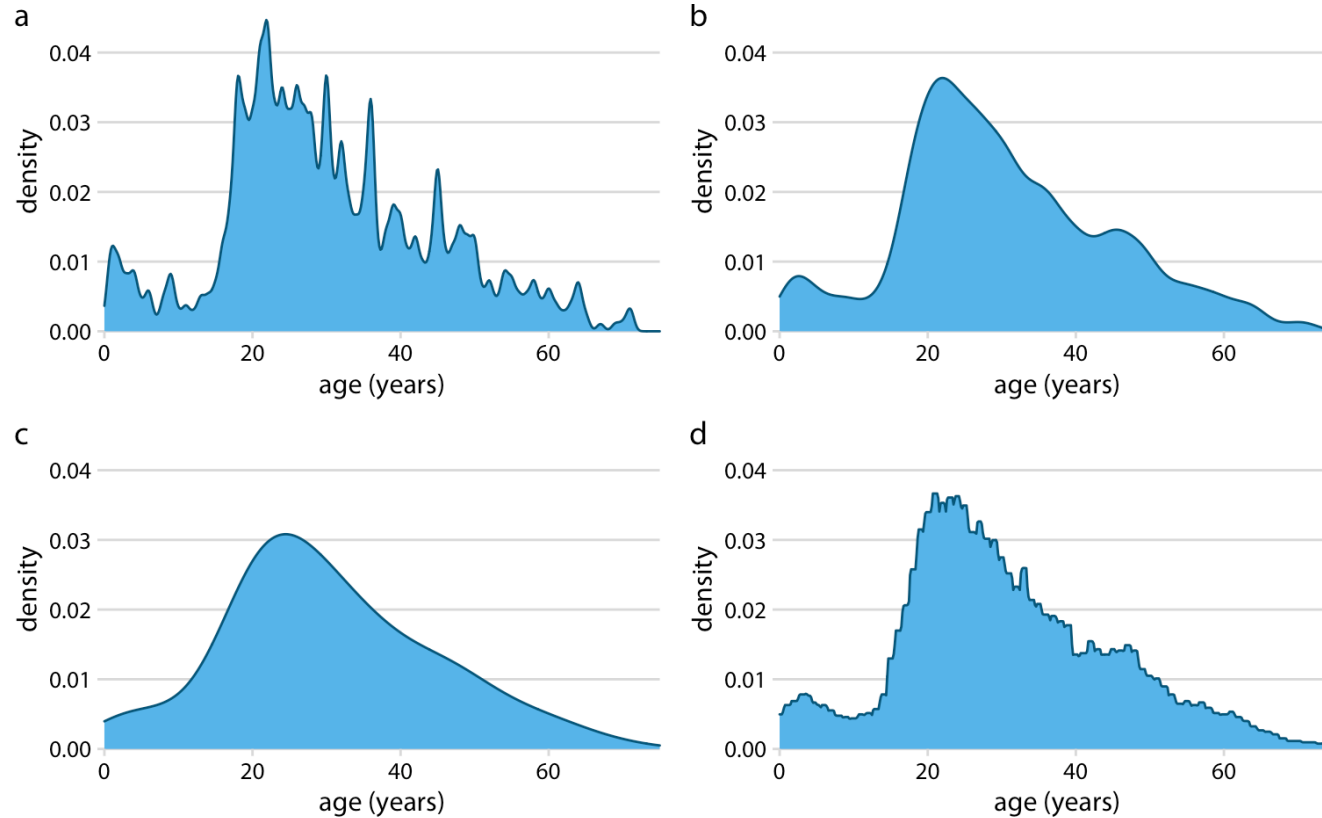


KDEs / Density Plots



Kernel density estimates depend on the chosen kernel and bandwidth. Here, the same age distribution of Titanic passengers is shown for four different combinations: (a) Gaussian kernel, bandwidth = 0.5; (b) Gaussian kernel, bandwidth = 2; (c) Gaussian kernel, bandwidth = 5; (d) Rectangular kernel, bandwidth = 2.

KDEs / Density Plots



- If the **bandwidth is too small**, then the density estimate can become overly peaky and visually busy and the **main trends in the data may be obscured**.
- If the **bandwidth is too large**, then smaller features in the **distribution of the data may disappear**. In addition, the choice of the kernel affects the shape of the density curve.

Where to use KDE Plot?

- Understanding data distribution (e.g., salary distribution).
- Comparing distributions (e.g., age distribution of men vs. women).
- Identifying skewness or multiple peaks (e.g., bimodal distributions).
- Density-based probability estimation (e.g., estimating likelihood of values).

Where to use KDE Plot?

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- Comparing distributions (e.g., age distribution of men vs. women).
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KDE Plot - Example

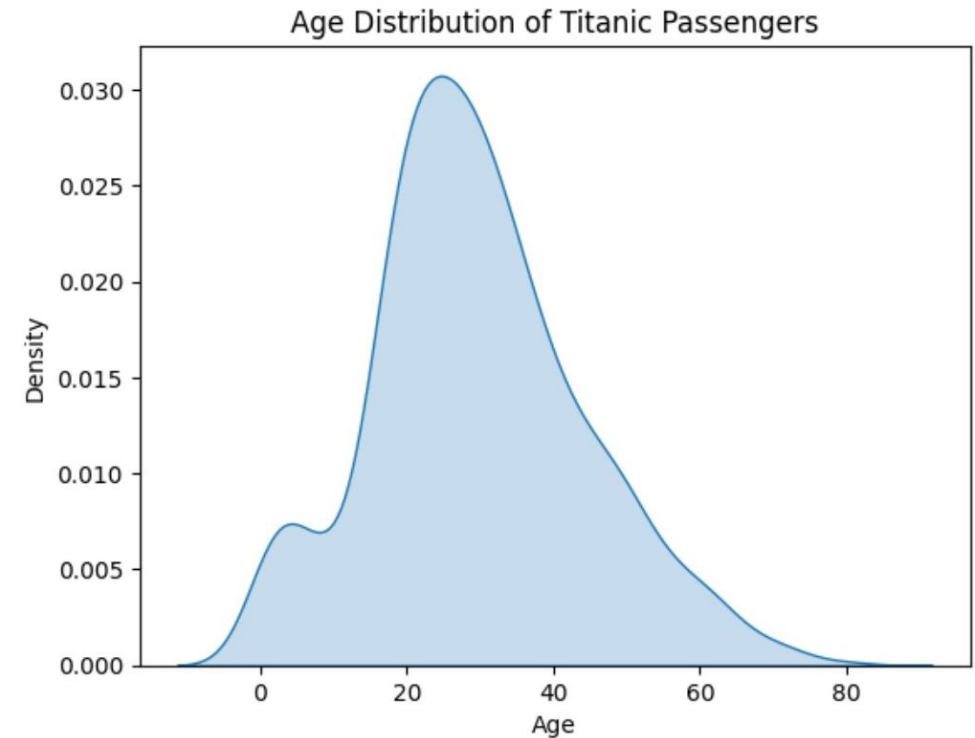
```
import seaborn as sns
import matplotlib.pyplot as plt

# Load the Titanic dataset
titanic = sns.load_dataset("titanic")

# Create KDE plot for age
sns.kdeplot(data=titanic, x="age", fill=True)

# Add labels and title
plt.xlabel("Age")
plt.ylabel("Density")
plt.title("Age Distribution of Titanic Passengers")

# Show the plot
plt.show()
```



KDE Plot - Example

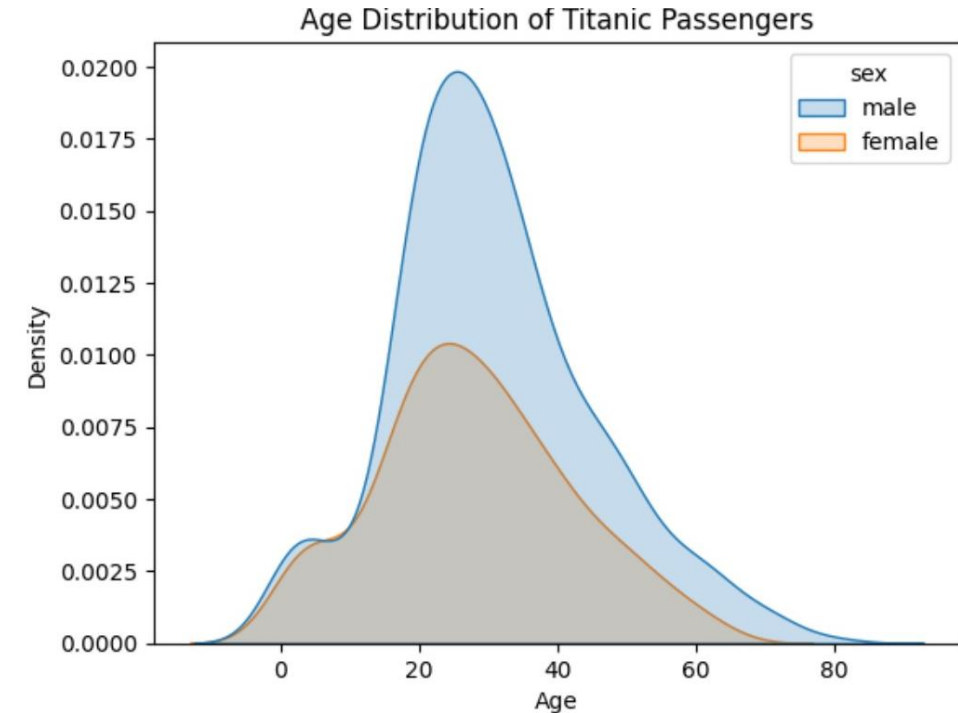
```
import seaborn as sns
import matplotlib.pyplot as plt

# Load the Titanic dataset
titanic = sns.load_dataset("titanic")

# Create KDE plot for age
sns.kdeplot(data=titanic, x="age", hue = 'sex', fill=True)

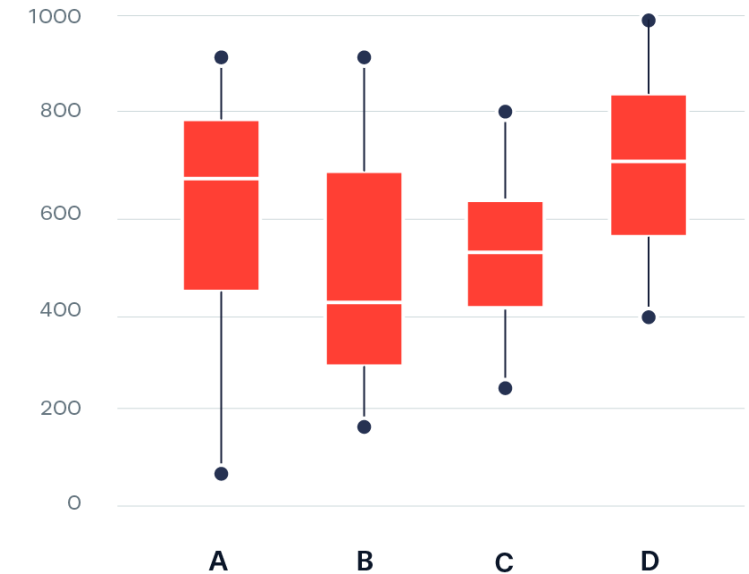
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plt.ylabel("Density")
plt.title("Age Distribution of Titanic Passengers")

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plt.show()
```



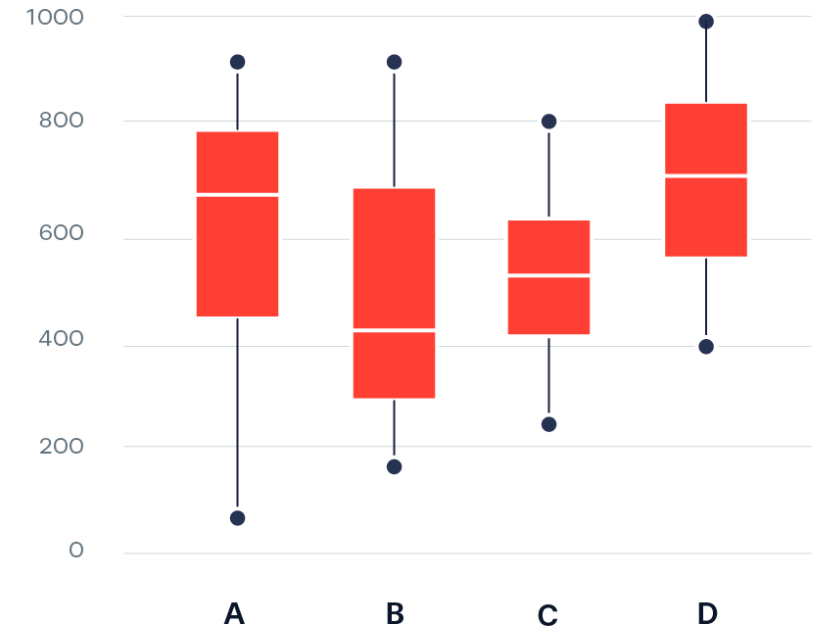
Box Plots

- A Box Plot is also known as a **Box and Whisker Plot**, and it is a graphical tool used to understand the **distribution of numerical data (Univariate)**.
- It shows the **median, quartiles and possible outliers** in a simple visual form.
- It helps compare **distributions between groups** with ease.

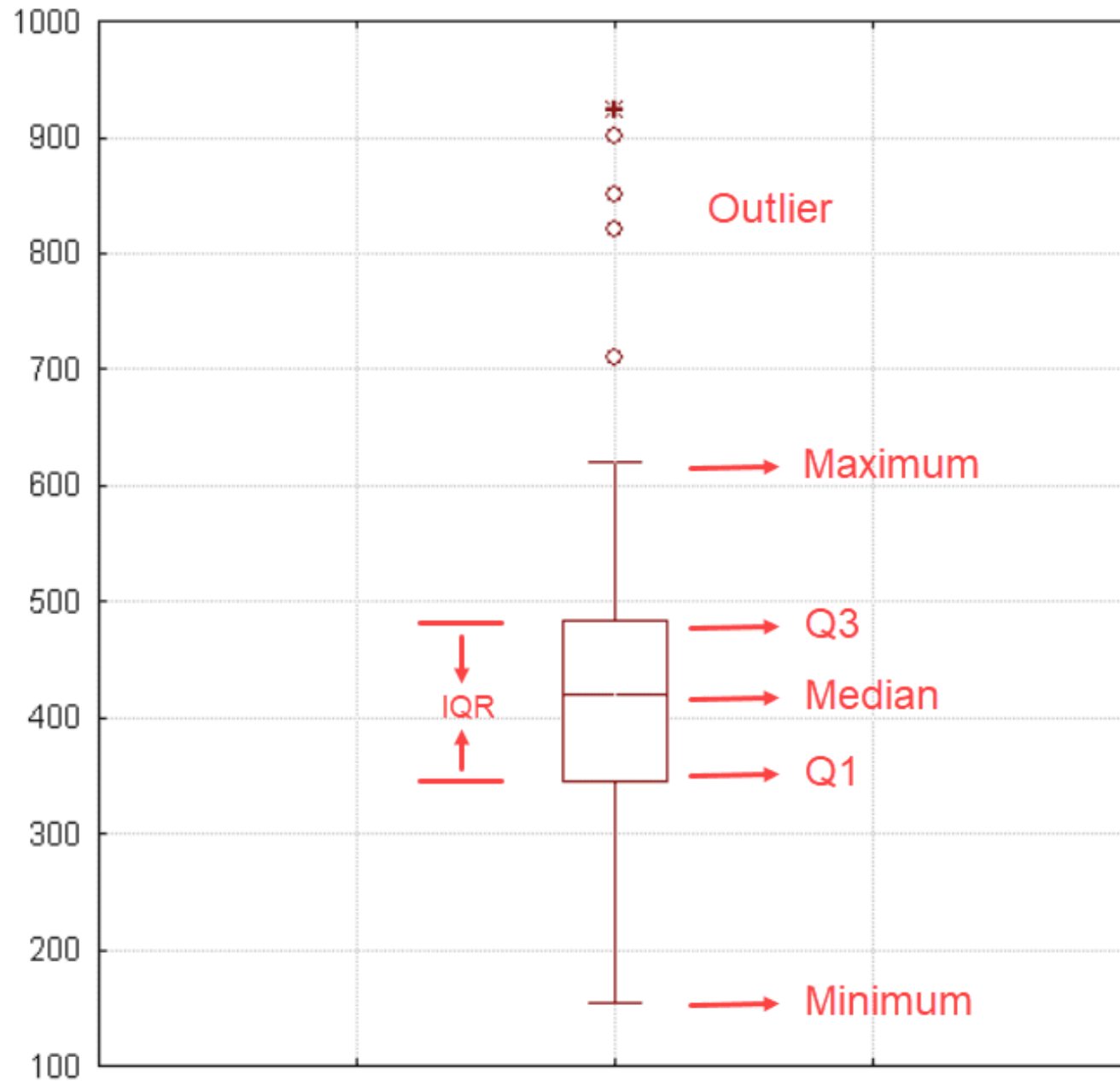


Box Plots

- The boxplot is primarily used to display the **distribution of a set of continuous data**.
- When you need to understand the data distribution characteristics or identify outliers, a boxplot is a useful tool for data analysis.
- Boxplots provide an intuitive view of the distribution of a dataset, **enabling quick identification of outliers**.

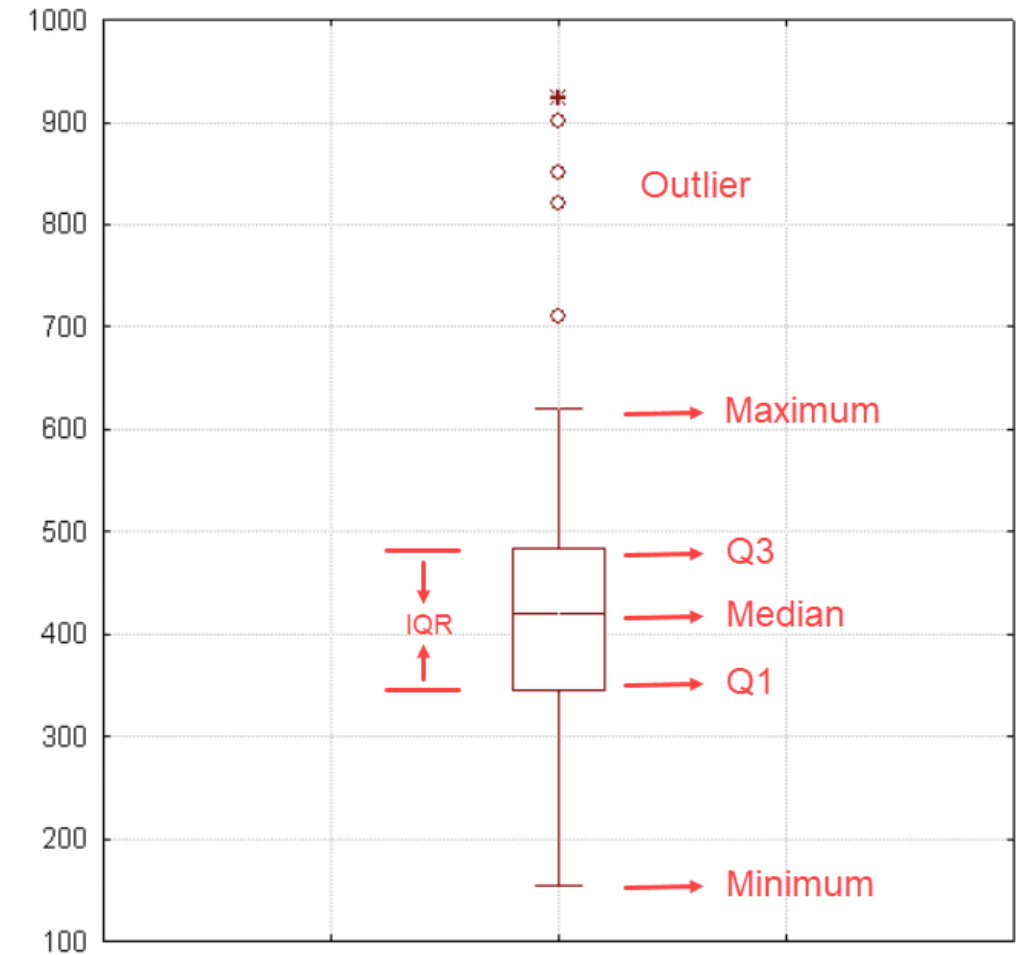


Box Plots



Box Plots

- **Minimum:** The smallest non-outlier value shown at the lower whisker end.
- **First Quartile (Q1):** The value below which 25% of the data lies.
- **Median (Q2):** The middle value that splits the data into two equal halves.
- **Third Quartile (Q3):** The value below which 75% of the data lies.
- **Maximum:** The largest non-outlier value shown at the upper whisker end.



Box Plots

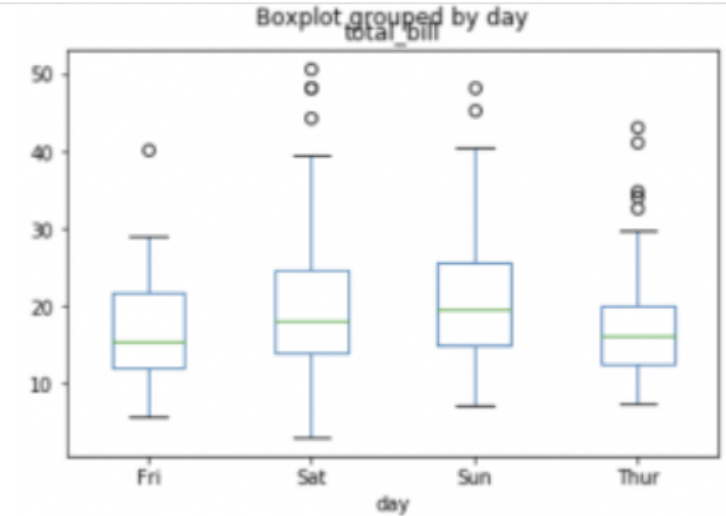
■ Box Plot with Seaborn

```
# import the required library
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
% matplotlib inline
```

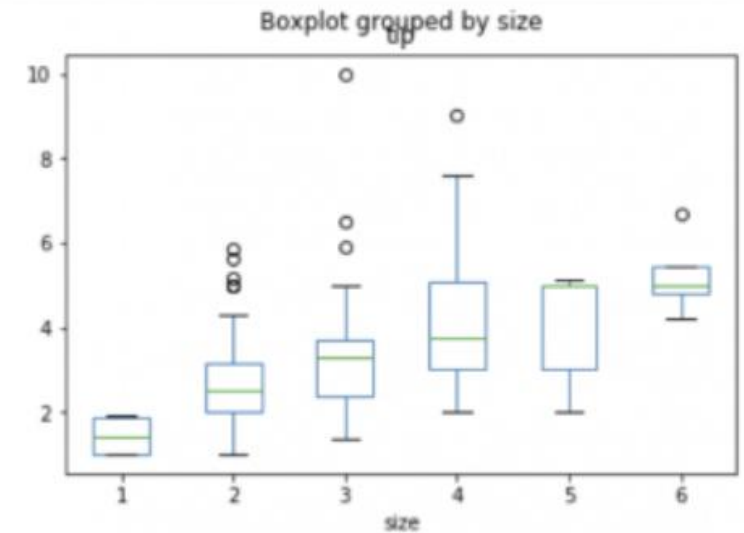
```
# Load the dataset
df = pd.read_csv("tips.csv")

# display 5 rows of dataset
df.head()
```

```
df.boxplot(by='day', column=['total_bill'], grid=False)
```



```
df.boxplot(by='size', column=['tip'], grid=False)
```



Box Plots

- [Box Plot with Seaborn](#)

Draw the boxplot using seaborn library:

Syntax :

```
seaborn.boxplot(x=None, y=None, hue=None, data=None, order=None, hue_order=None,  
orient=None, color=None, palette=None, saturation=0.75, width=0.8, dodge=True,  
fliersize=5, linewidth=None, whis=1.5, notch=False, ax=None, **kwargs)
```

Parameters:

x = feature of dataset

y = feature of dataset

hue = feature of dataset

data = dataframe or full dataset

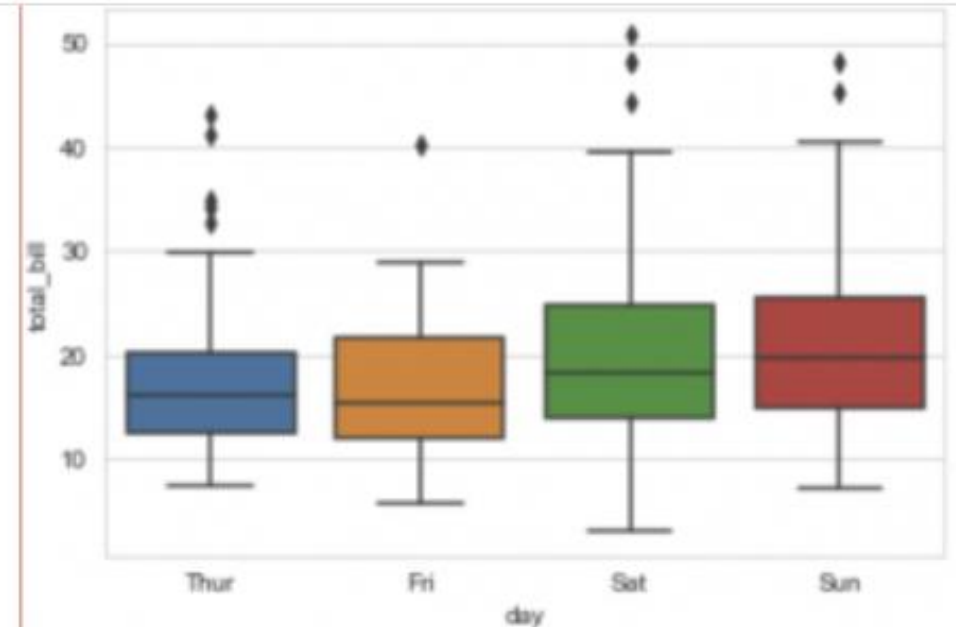
color = color name

Box Plots

- Box Plot with Seaborn

```
# Draw a vertical boxplot grouped
# by a categorical variable:
sns.set_style("whitegrid")
```

```
sns.boxplot(x = 'day', y = 'total_bill', data = tips)
```



Box Plots

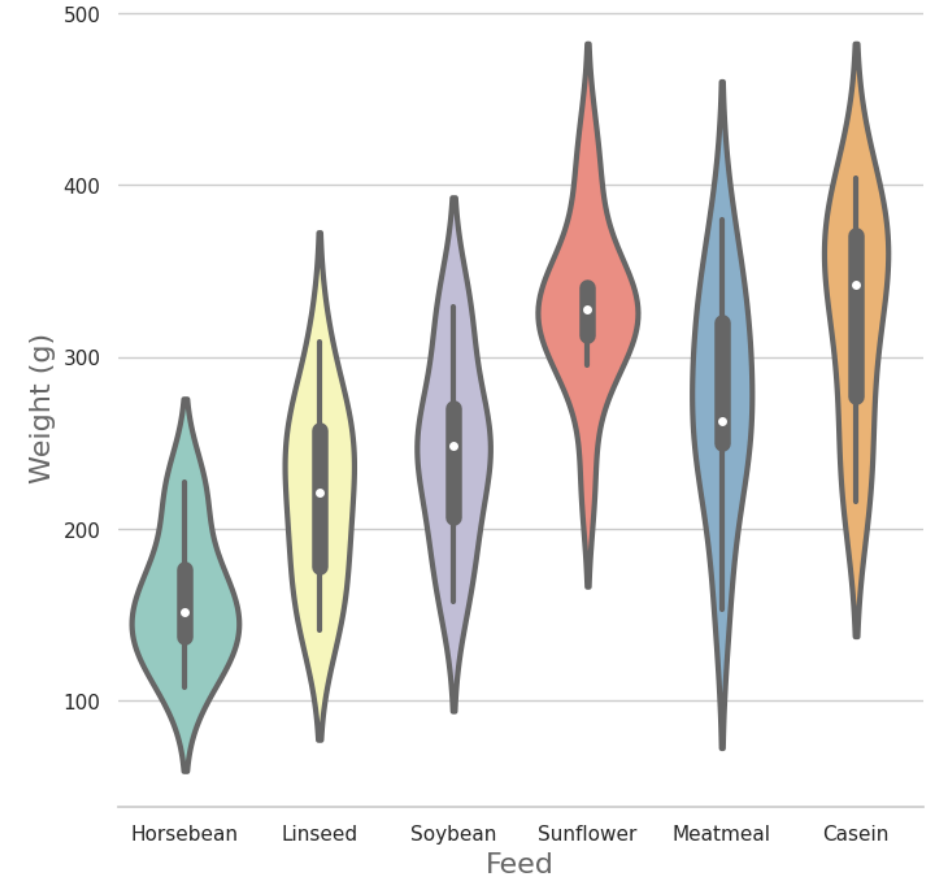
LEARNING
IS
AN
ENRICHING
EXPERIENCE

Feature	Vertical Box Plot	Horizontal Box Plot
Orientation	Box is drawn vertically along the y-axis	Box is drawn horizontally along the x-axis
Best For	Comparing numerical values across categories.	Showing distribution when category labels are long.
Readability	Can become cramped with many categories.	More readable when dataset has many labels.
Common Use	Default plot in most statistical tools.	Useful for survey data, ratings and long text labels.
Space Utilization	Requires more vertical space.	Utilizes horizontal space more efficiently.

Violin Plots

- A violin plot is a **hybrid of a box plot and a kernel density plot**, which shows peaks in the data.
- It is used to visualize the distribution of numerical data.
- Unlike a box plot that can only show summary statistics, violin plots depict **summary statistics and the density of each variable**.

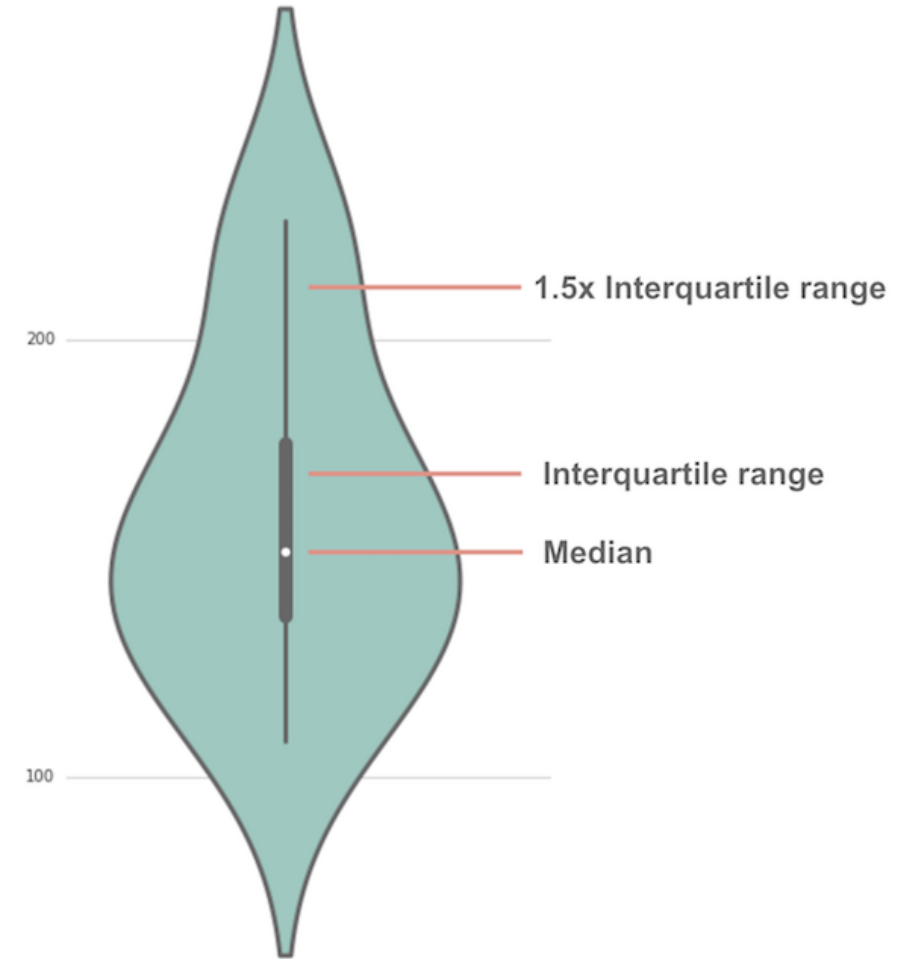
Chick weights by feed type



How to read a Violin Plot

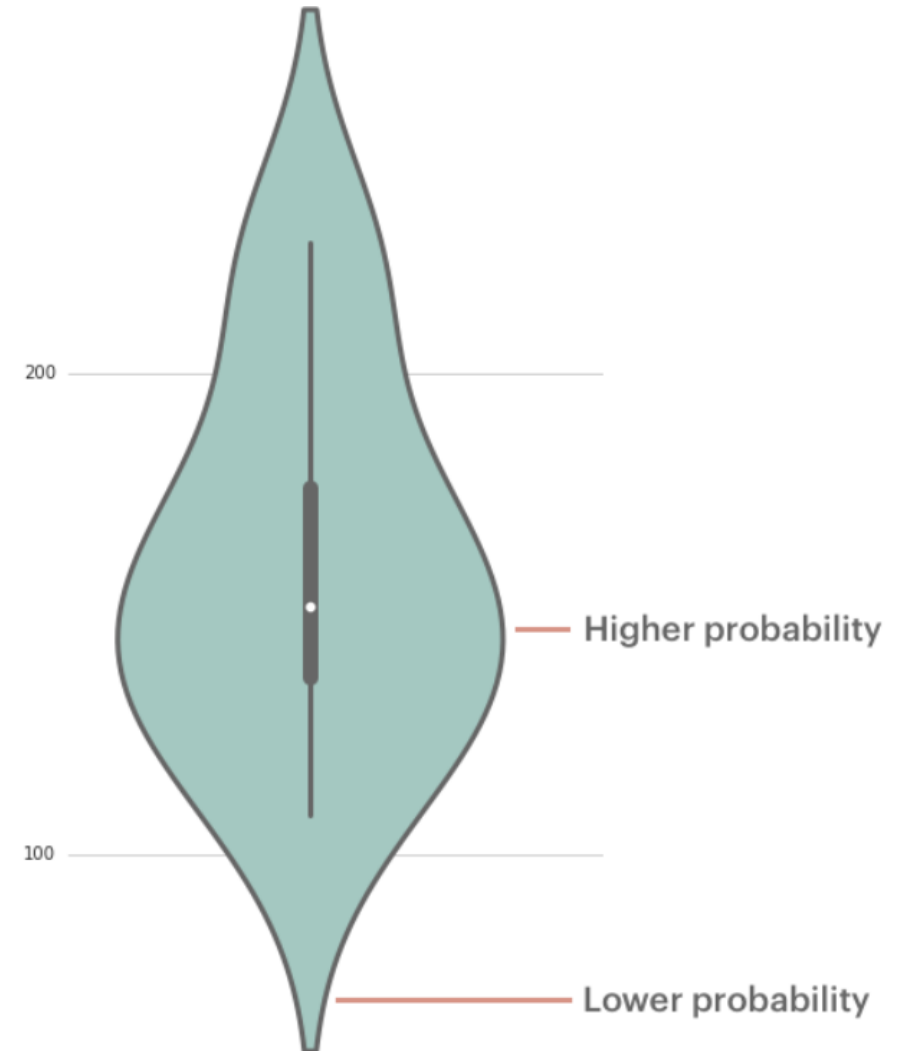
Violin plots have many of the same summary statistics as box plots:

- the **white dot** represents the **median**.
- the **thick gray bar** in the center represents the **interquartile range**.
- the thin gray line represents the rest of the distribution, except for points that are determined to be “outliers” using a method that is a function of the interquartile range.



How to read a Violin Plot

- On each side of the gray line is a kernel density estimation to show the distribution shape of the data.
- **Wider sections** of the violin plot represent a **higher probability** that members of the population will take on the given value; the **skinnier sections** represent a **lower probability**.



Violin Plot

```
import matplotlib.pyplot as plt
import numpy as np
```

```
# Sample data
```

```
np.random.seed(42)
```

```
group_a = np.random.normal(50, 10, 100)
```

```
group_b = np.random.normal(60, 8, 100)
```

```
group_c = np.random.normal(55, 12, 100)
```

```
data = [group_a, group_b, group_c]
```

```
# Create violin plot
```

```
plt.figure()
```

```
plt.violinplot(data, showmeans=True)
```

```
plt.xticks([1, 2, 3], ['Group A', 'Group B', 'Group C'])
```

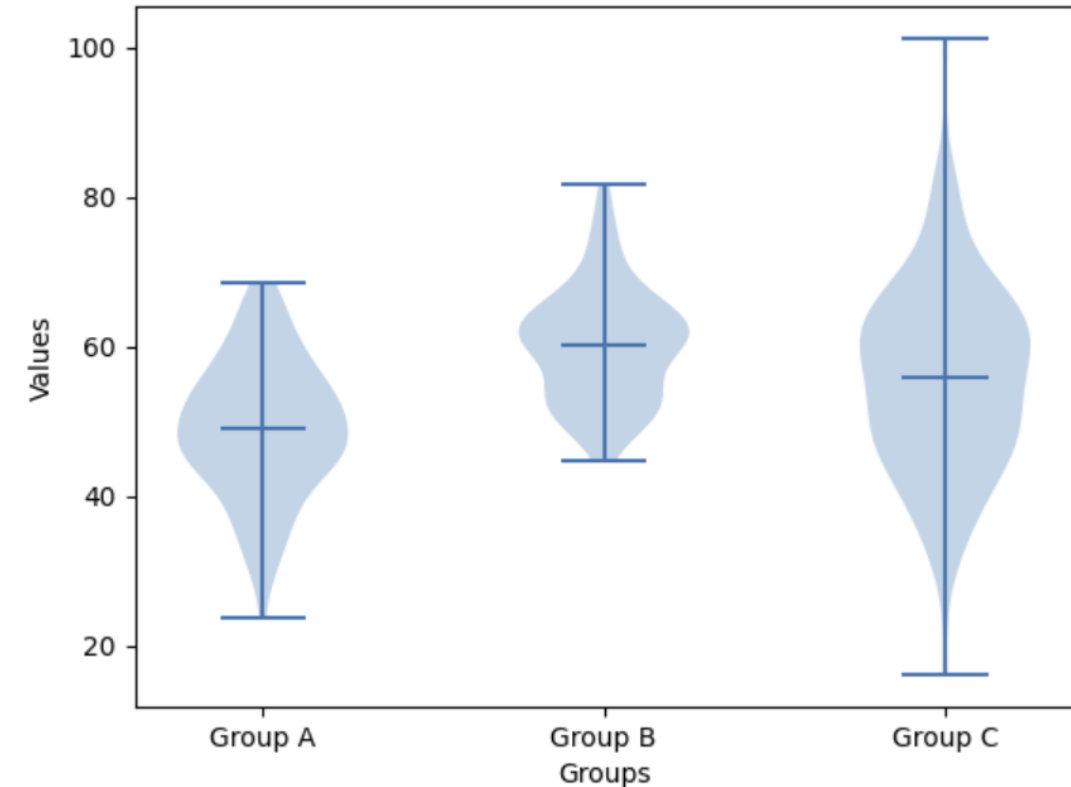
```
plt.xlabel('Groups')
```

```
plt.ylabel('Values')
```

```
plt.title('Simple Violin Plot Example')
```

```
plt.show()
```

Simple Violin Plot Example



Violin Plot

```
np.random.normal(mean, std, size)
```

This function generates **random values from a normal (Gaussian) distribution**.

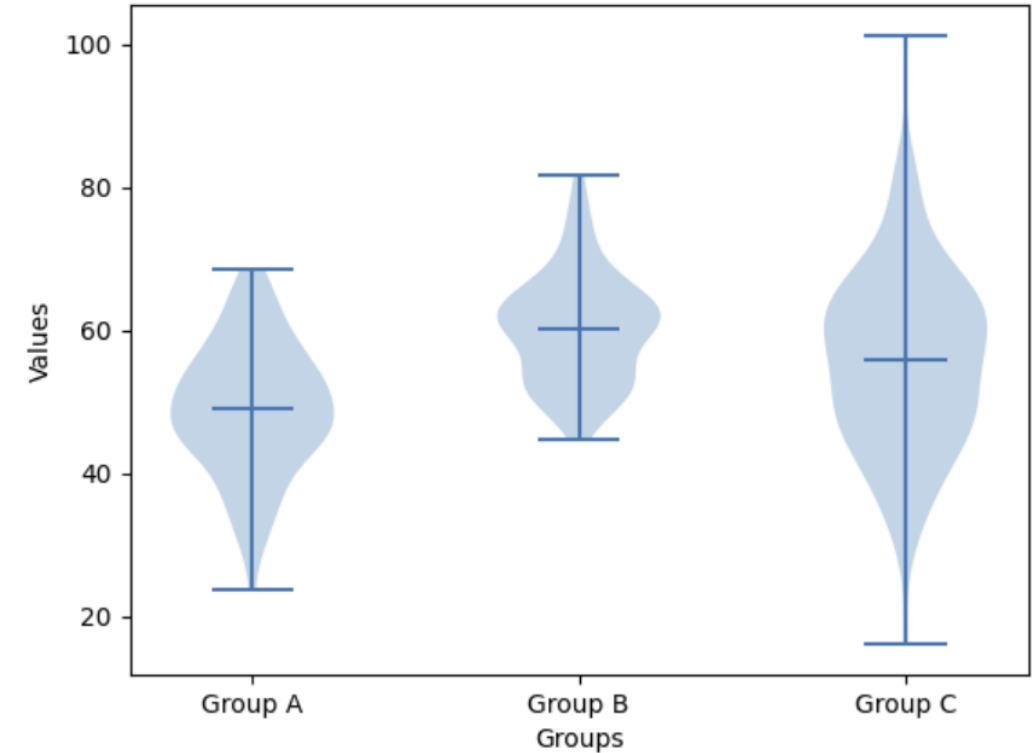
```
group_a = np.random.normal(50, 10, 100)
```

- Mean = 50
- Standard deviation = 10
- Samples = 100

Interpretation:

100 data points centered around 50, with moderate spread.

Simple Violin Plot Example



[Interactive Python File - Click Here](#)

Comparison Table

Plot Type	Univariate	Bivariate	Multivariate	Typical Use
Histogram	✓ Yes	⚠ Limited	✗ No	Distribution shape
Box Plot	✓ Yes	✓ Yes	✓ Yes	Spread & outliers
KDE	✓ Yes	✓ Yes	✗ Rare	Density estimation
Violin Plot	✓ Yes	✓ Yes	✓ Yes	Distribution comparison

LEARNING
AUGMENTED

Exploratory Data Analysis (EDA)

<https://www.geeksforgeeks.org/data-analysis/what-is-exploratory-data-analysis/>

<https://github.com/swarnava-96/Exploratory-Data-Analysis-EDA/blob/main/EDA3.ipynb>

<https://github.com/Sukanyasingh3/Exploratory-data-analysis/tree/main/EDA>

<https://www.geeksforgeeks.org/data-visualization/violin-plot-for-data-analysis/>

References

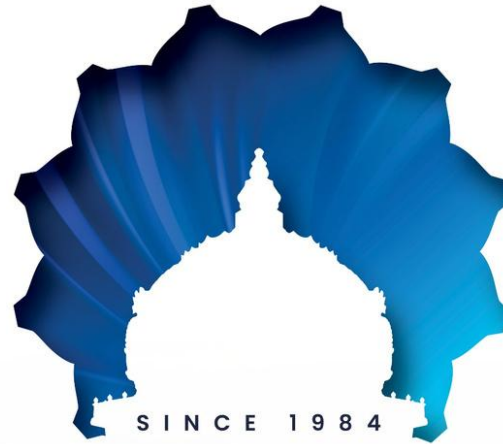
<https://www.analyticsvidhya.com/blog/2021/08/exploratory-data-analysis-and-visualization-techniques-in-data-science/>

<https://www.geeksforgeeks.org/data-analysis/eda-with-numpy-pandas-matplotlib-seaborn/>

<https://www.geeksforgeeks.org/data-visualization/plotting-histogram-in-python-using-matplotlib/>

<https://clauswilke.com/dataviz/histograms-density-plots.html>

<https://www.geeksforgeeks.org/machine-learning/box-plot/>



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